PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 1 · Version A

21 questions · 90 minutes · original practice exam, not an official exam

Name _____ NetID _____ Section _____

Closed book. Calculators allowed. Use the official PHY 2111 formula sheet. Take $g = 9.81\ m/s^2$.
MC5 questions have five choices (6 points), MC3 three choices (3 points), MC2 two choices (2 points).
Mark one answer per question. Answers and worked solutions are in the separate solutions file.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A car starts from rest and accelerates uniformly at $2.5\ m/s^2$ for 6 s. It then moves at constant speed for 10 s, and finally brakes uniformly to a stop in 5 s.

1) What is the maximum speed of the car? **MC5**

- **a.** 12.5 m/s
- **b.** 30 m/s
- **c.** 37.5 m/s
- **d.** 7.5 m/s
- **e.** 15 m/s

2) What total distance does the car travel? **MC5**

- **a.** 315 m
- **b.** 150 m
- **c.** 233 m
- **d.** 195 m
- **e.** 270 m

3) While the car is braking, its acceleration vector points **MC3**

- **a.** opposite to the direction of motion.
- **b.** in the direction of motion.
- **c.** nowhere: the acceleration is zero.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A soccer ball is kicked from level ground with an initial speed of 18 m/s at 40° above the horizontal. Ignore air resistance.

4) How long is the ball in the air? **MC5**

- **a.** 1.18 s
- **b.** 2.36 s
- **c.** 4.72 s
- **d.** 3.67 s
- **e.** 1.41 s

5) How far from the kicker does the ball land? **MC5**

- **a.** 27.3 m
- **b.** 32.5 m
- **c.** 42.5 m
- **d.** 33 m
- **e.** 16.3 m

6) What is the speed of the ball at the top of its path? **MC3**

- **a.** 0 m/s
- **b.** 13.8 m/s
- **c.** 18 m/s

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A stone is thrown horizontally at 8 m/s from the top of a 25 m tall cliff.

7) How far from the base of the cliff does the stone land? **MC5**

- **a.** 12.8 m
- **b.** 9.03 m
- **c.** 18.1 m
- **d.** 25 m
- **e.** 36.1 m

8) If the stone were thrown horizontally at 16 m/s instead, the time it takes to reach the ground would **MC3**

- **a.** stay the same.
- **b.** be cut in half.
- **c.** double.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A river 80 m wide flows east at 1.5 m/s. A boat moves at 4 m/s relative to the water.

9) The boat points due north. How far downstream does it land? MC5

- **a.** 30 m
- **b.** 0 m
- **c.** 12 m
- **d.** 20 m
- **e.** 48 m

10) At what angle west of north must the boat point to land directly across? MC5

- **a.** 26.6 °
- **b.** 22 °
- **c.** 68 °
- **d.** 0 °
- **e.** 20.6 °

11) Compared with pointing due north, the crossing time when the boat lands directly across is MC3

- **a.** shorter.
- **b.** the same.
- **c.** longer.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 4 kg block sits on a frictionless incline at 30°. It is tied by a light string over a frictionless pulley at the top of the incline to a 3 kg block hanging vertically. The blocks are released from rest.

12) What is the magnitude of the acceleration of the blocks? MC5

- **a.** 1.4 m/s²
- **b.** 2.8 m/s²
- **c.** 7.01 m/s²
- **d.** 4.2 m/s²
- **e.** 0 m/s²

13) What is the tension in the string? MC5

- **a.** 29.4 N
- **b.** 19.6 N
- **c.** 25.2 N
- **d.** 5.61 N
- **e.** 33.6 N

14) Compared with the weight of the hanging block, the tension is **MC3**

a. smaller.
b. equal.
c. larger.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 12 kg box is dragged across a level floor at constant velocity by a rope pulling at 25° above the horizontal. The coefficients of friction are $\mu_s = 0.5$ and $\mu_k = 0.35$.

15) What is the tension in the rope? **MC5**

a. 41.2 N
b. 58.9 N
c. 37.3 N
d. 39.1 N
e. 45.5 N

16) The normal force on the box is **MC3**

a. greater than mg.
b. less than mg.
c. equal to mg.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 400 kg roller-coaster car travels upside down at the top of a vertical loop of radius 12 m with a speed of 14 m/s. The car rides on the inside of the track.

17) What is the magnitude of the normal force from the track on the car at the top? **MC5**

a. 1.05×10^4 N
b. 2610 N
c. 6530 N
d. 0 N
e. 3920 N

18) What is the minimum speed at the top for the car to stay in contact with the track? **MC5**

a. 24.3 m/s
b. 10.8 m/s
c. 23.5 m/s
d. 15.3 m/s
e. 5.42 m/s

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A satellite moves in a circular orbit 2000 km above Earth's surface. ($R_E = 6.37 \times 10^6$ m, $M_E = 5.97 \times 10^{24}$ kg, $G = 6.67 \times 10^{-11}$ N·m²/kg²)

19) What is the orbital speed of the satellite? **MC5**

- **a.** 3.45 km/s
- **b.** 6.9 km/s
- **c.** 4.76 km/s
- **d.** 9.75 km/s
- **e.** 7.91 km/s

20) What is the period of the orbit? **MC5**

- **a.** 2.12 h
- **b.** 24 h
- **c.** 4.24 h
- **d.** 1.06 h
- **e.** 3 h

21) If the satellite had twice the mass, its orbital speed at the same radius would be **MC3**

- **a.** the same.
- **b.** half as large.
- **c.** twice as large.